

$I = \int_a^b f(x) \, \mathrm{d}x$
 $f(x)$
 $F(x)$

$\int_a^b f(x) \, \mathrm{d}x = F(b) - F(a).$

$\frac{\sin x}{x}$
 $\sin x^2$
 $f(x)$
 (a,b)
 ξ

$\int_a^b f(x) \, \mathrm{d}x = (b-a)f(\xi).$

$b-a$
 $f(\xi)$
 I
 ξ
 $f(\xi)$
 $f(\xi)$
 $[a,b]$
 $f(\xi)$
 $f(x)$
 a
 b
 $f(\xi)$
 x
 y
 O
 a
 ξ
 b
 $f(x)$
 $f(\xi)$

$f(a)$
 $f(b)$
 $f(\xi)$

$$T = \frac{b-a}{2}[f(a) + f(b)] \tag{4.1.1}$$

$$c = \frac{a+b}{2}$$

$f(c)$
 $f(\xi)$

$$R = (b-a)f\left(\frac{a+b}{2}\right). \tag{4.1.2}$$

$f(x)$
 a
 b
 x
 y
 O
 a
 b
 $f(x)$
 $f(a)$
 $f(b)$
 $[a,b]$
 x_k
 $f(x_k)$
 $f(\xi)$

$$\int_a^b f(x) \, \mathrm{d}x \approx \sum_{k=0}^n A_k f(x_k). \tag{4.1.3}$$

x_k
 A_k
 x_k
 A_k
 x_k

$$\begin{array}{l} f(x) \\ m \\ m+1 \\ m \\ m \\ f(x)=1,x,x^2,\cdots,x^m \end{array}$$

$$\left\{ \begin{array}{l} \sum A_k=b-a, \\ \sum A_kx_k=\frac{1}{2}(b^2-a^2), \\ \vdots \\ \sum A_kx_k^m=\frac{1}{m+1}(b^{m+1}-a^{m+1}). \end{array} \right. \tag{4.1.4}$$

$$\begin{array}{l} \sum\limits_{k=0}^n \\ x_k \\ [a,b] \\ m=n \\ A_k \\ n \\ x_k \\ A_k \end{array}$$

$$a\leq x_0<x_1<x_2<\cdots <x_n\leq b$$

$$\begin{array}{l} f(x) \\ L_n(x) \\ L_n(x) \\ I_n=\int_a^b L_n(x)\,\mathrm{d}x \\ I=\int_a^b f(x)\,\mathrm{d}x \end{array}$$

$$I_n=\sum_{k=0}^n A_k f(x_k) \tag{4.1.5}$$

$$\begin{array}{l} A_k \\ l_k(x) \end{array}$$

$$A_k=\int_a^bl_k(x)\,\mathrm{d}x\tag{4.1.6}$$

$$R[f]=I-I_n=\int_a^b\frac{f^{n+1}(\xi)}{(n+1)!}w(x)\,\mathrm{d}x,\tag{4.1.7}$$

ξ

x

$$w(x)=(x-x_0)(x-x_1)\cdots(x-x_n)$$

n

$$f(x)$$

$$R[f]$$

n

n

$$l_k(x)$$

$$\int_a^b l_k(x)\,\mathrm{d}x=\sum_{j=0}^n A_j l_k(x_j).$$

$$l_k(x_j)=\delta_{kj}$$

$$A_k$$

n

$$f^{n+1}(\xi)$$

$$n+1$$

$$[a,b]$$

n

$$h=\frac{b-a}{n}$$

$$x_k=a+kh$$

$$I_n=(b-a)\sum_{k=0}^nC_k^{(n)}f(x_k)\tag{4.2.1}$$

$$C_k^{(n)}$$

$$x=a+th,$$

$$C_k^{(n)}=\frac{h}{b-a}\int_0^n\prod_{\substack{j=0\\j\neq k}}^n\frac{t-j}{k-j}\,\mathrm{d}t=\frac{(-1)^{n-k}}{nk!(n-k)!}\int_0^n\prod_{\substack{j=0\\j\neq k}}^n(t-j)\,\mathrm{d}t.\tag{4.2.2}$$

$$n = 1$$

$$C_0^{(1)} = C_1^{(1)} = \frac{1}{2},$$

$$n = 2$$

$$\begin{aligned} C_0^{(2)} &= \frac{1}{4} \int_0^2 (t-1)(t-2) \, dt = \frac{1}{6}, \\ C_1^{(2)} &= -\frac{1}{2} \int_0^2 t(t-2) \, dt = \frac{4}{6}, \quad C_2^{(2)} = \frac{1}{4} \int_0^2 t(t-1) \, dt = \frac{1}{6}. \\ S &= \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]. \end{aligned} \quad (4.2.3)$$

$$n = 4$$

$$C = \frac{b-a}{90} [7f(x_0) + 32f(x_1) + 12f(x_2) + 32f(x_3) + 7f(x_4)]. \quad (4.2.4)$$

$$\begin{aligned} x_k &= a + kh \\ h &= \frac{b-a}{4} \end{aligned}$$

$$n$$

$$C_0^{(n)}$$

$$C_1^{(n)}$$

$$C_2^{(n)}$$

$$C_3^{(n)}$$

$$C_4^{(n)}$$

$$C_5^{(n)}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{6}$$

$$\frac{2}{3}$$

$$\frac{1}{6}$$

$$\frac{1}{8}$$

$$\frac{3}{8}$$

$$\frac{3}{8}$$

$$\frac{1}{8}$$

$$\frac{7}{90}$$

$$\begin{array}{c}
\frac{16}{45} \\
\frac{2}{15} \\
\frac{16}{45} \\
\frac{7}{90} \\
\frac{19}{288} \\
\frac{25}{96} \\
\frac{25}{144} \\
\frac{25}{144} \\
\frac{25}{96} \\
\frac{19}{288} \\
n \\
C_0^{(n)} \\
C_1^{(n)} \\
C_2^{(n)} \\
C_3^{(n)} \\
C_4^{(n)} \\
C_5^{(n)} \\
C_6^{(n)} \\
C_7^{(n)} \\
C_8^{(n)} \\
\frac{41}{840} \\
\frac{9}{35} \\
\frac{9}{280} \\
\frac{34}{105} \\
\frac{9}{280} \\
\frac{9}{35} \\
\frac{41}{840} \\
\frac{751}{17280} \\
\frac{3577}{17280} \\
\frac{1323}{17280} \\
\frac{2989}{17280} \\
\frac{2989}{17280} \\
\frac{1323}{17280}
\end{array}$$

$$\begin{array}{r} \frac{3577}{17280} \\ \frac{751}{17280} \\ \frac{989}{28350} \\ \frac{5888}{28350} \\ \frac{-928}{28350} \\ \frac{10496}{28350} \\ \frac{-4540}{28350} \\ \frac{10496}{28350} \\ \frac{-928}{28350} \\ \frac{5888}{28350} \\ \frac{989}{28350} \\ n \geq 8 \\ n \geq 8 \\ n \\ n = 9 \end{array}$$

$$\left(\frac{2857}{89600}, \frac{15741}{89600}, \frac{27}{2240}, \frac{1209}{5600}, \frac{2889}{44800}, \frac{2889}{44800}, \frac{1209}{5600}, \frac{27}{2240}, \frac{15741}{89600}, \frac{2857}{89600}\right),$$

$n = 8$

n

n

$f(x) = x^3$

$$S = \frac{b-a}{6}\left[a^3 + 4\left(\frac{a+b}{2}\right)^3 + b^3 \right].$$

$I = \int_a^b x^3 \, \mathrm{d}x = \frac{b^4 - a^4}{4}$

$S = I$

$f(x) = x^4$

n

$n+1$

n

$f(x) = x^{n+1}$

$f^{(n+1)}(x) = (n+1)!$

$$R[f]=\int_a^b\prod_{j=0}^n(x-x_j)\,\mathrm{d}x.$$

$$\begin{array}{l}x=a+th\\x_j=a+jh\end{array}$$

$$R[f]=h^{n+2}\int_0^n\prod_{j=0}^n(t-j)\,\mathrm{d}t.$$

$$\begin{array}{l}n\\[2mm]\frac{n}{2}\\[2mm]t=u+\frac{n}{2}\end{array}$$

$$R[f]=h^{n+2}\int_{-n/2}^{n/2}\prod_{j=0}^n\Big(u+\frac{n}{2}-j\Big)\,\mathrm{d}u,$$

$$R[f]=0$$

$$H(u)=\prod_{j=0}^n\Big(u+\frac{n}{2}-j\Big)=\prod_{j=-n/2}^{n/2}(u-j)$$

$$R_{\mathrm{T}}=I-T=\int_a^b\frac{f''(\xi)}{2}(x-a)(x-b)\,\mathrm{d}x.$$

$$\begin{array}{l}(x-a)(x-b)\\[2mm][a,b]\\[2mm](a,b)\\[2mm]\eta\end{array}$$

$$R_{\mathrm{T}}=\frac{f''(\eta)}{2}\int_a^b(x-a)(x-b)\,\mathrm{d}x=-\frac{f''(\eta)}{12}(b-a)^3,\quad \eta\in(a,b). \quad (4.2.5)$$

$$\begin{array}{l}R_{\mathrm{S}}=I-S\\[2mm]H(x)\end{array}$$

$$H(a)=f(a),\quad H(b)=f(b),\quad H(c)=f(c),\quad H'(c)=f'(c),\quad (4.2.6)$$

$$c=\frac{a+b}{2}$$

$$H(x)$$

$$\int_a^b H(x) \, dx = \frac{b-a}{6} [H(a) + 4H(c) + H(b)].$$

$$S$$

$$R_S = I - S = \int_a^b [f(x) - H(x)] \, dx.$$

$$H(x)$$

$$f(x) - H(x) = \frac{f^{(4)}(\xi)}{4!} (x-a)(x-c)^2(x-b),$$

$$R_S = \int_a^b \frac{f^{(4)}(\xi)}{4!} (x-a)(x-c)^2(x-b) \, dx.$$

$$(x-a)(x-c)^2(x-b)$$

$$[a,b]$$

$$R_S = \frac{f^{(4)}(\eta)}{4!} \int_a^b (x-a)(x-c)^2(x-b) \, dx = -\frac{b-a}{180} \left(\frac{b-a}{2} \right)^4 f^{(4)}(\eta). \quad (4.2.7)$$

$$R_C = I - C = -\frac{2(b-a)}{945} \left(\frac{b-a}{4} \right)^6 f^{(6)}(\eta). \quad (4.2.8)$$

$$[a,b]$$

$$n$$

$$h = \frac{b-a}{n}$$

$$x_k = a + kh$$

$$k = 0, 1, \dots, n$$

$$[x_k, x_{k+1}]$$

$$I_k$$

$$n-1$$

$$\sum_{k=0}^{n-1} I_k$$

$$I$$

$$T_n = \sum_{k=0}^{n-1} \frac{h}{2} [f(x_k) + f(x_{k+1})] = \frac{h}{2} \left[f(a) + 2 \sum_{k=1}^{n-1} f(x_k) + f(b) \right], \quad (4.2.9)$$

$$I - T_n = \sum_{k=0}^{n-1} \left[-\frac{h^3}{12} f''(\eta_k) \right] = -\frac{b-a}{12} h^2 f''(\eta). \quad (4.2.10)$$

$$\begin{aligned} &[x_k, x_{k+1}] \\ &x_{k+\frac{1}{2}} \end{aligned}$$

$$\begin{aligned} S_n &= \sum_{k=0}^{n-1} \frac{h}{6} [f(x_k) + 4f(x_{k+\frac{1}{2}}) + f(x_{k+1})] \\ &= \frac{h}{6} \left[f(a) + 4 \sum_{k=0}^{n-1} f(x_{k+\frac{1}{2}}) + 2 \sum_{k=1}^{n-1} f(x_k) + f(b) \right]. \end{aligned} \quad (4.2.11)$$

$$\begin{aligned} &[x_k, x_{k+1}] \\ &x_{k+\frac{1}{4}} \\ &x_{k+\frac{1}{2}} \\ &x_{k+\frac{3}{4}} \end{aligned}$$

$$\begin{aligned} C_n &= \frac{h}{90} \left[7f(a) + 32 \sum_{k=0}^{n-1} f(x_{k+\frac{1}{4}}) + 12 \sum_{k=0}^{n-1} f(x_{k+\frac{1}{2}}) \right. \\ &\quad \left. + 32 \sum_{k=0}^{n-1} f(x_{k+\frac{3}{4}}) + 14 \sum_{k=1}^{n-1} f(x_k) + 7f(b) \right]. \end{aligned} \quad (4.2.12)$$

$$I - S_n = -\frac{b-a}{180} \left(\frac{h}{2} \right)^4 f^{(4)}(\eta), \quad \eta \in [a, b], \quad (4.2.13)$$

$$I - C_n = -\frac{2(b-a)}{945} \left(\frac{h}{4} \right)^6 f^{(6)}(\eta), \quad \eta \in [a, b]. \quad (4.2.14)$$

$$\begin{aligned} f(x) &= \frac{\sin x}{x} \\ I &= \int_0^1 \frac{\sin x}{x} dx \\ [0, 1] \\ T_8 &= 0.945\,690\,9 \\ [0, 1] \\ S_4 &= 0.946\,083\,2 \\ T_8 \\ S_4 \end{aligned}$$

$$I = 0.946\,083\,1$$

$$T_8 = 0.945\,690\,9$$

$$S_4 = 0.946\,083\,2$$

$$|I - T_8| = 0.0003922 < 0.0005$$

$$0.00005$$

$$T_8$$

$$h \rightarrow 0$$

$$I$$

$$h$$

$$x$$

$$f(x)$$

$$0$$

$$1$$

$$1/8$$

$$0.997\,397\,8$$

$$1/4$$

$$0.989\,615\,8$$

$$3/8$$

$$0.976\,726\,7$$

$$1/2$$

$$0.958\,851\,0$$

$$5/8$$

$$0.936\,155\,6$$

$$3/4$$

$$0.908\,851\,6$$

$$7/8$$

$$0.877\,192\,5$$

$$1$$

$$0.841\,470\,9$$

$$\frac{I - T_n}{h^2} = -\frac{1}{12} \sum_{k=0}^{n-1} h f''(\eta_k) \rightarrow -\frac{1}{12} \int_a^b f''(x) \, dx,$$

$$h \rightarrow 0$$

$$\begin{aligned}\frac{I-T_n}{h^2} &\rightarrow -\frac{1}{12}[f'(b)-f'(a)], \\ \frac{I-S_n}{h^4} &\rightarrow -\frac{1}{180\times 2^4}[f'''(b)-f'''(a)], \\ \frac{I-C_n}{h^6} &\rightarrow -\frac{2}{945\times 4^6}[f^{(5)}(b)-f^{(5)}(a)].\end{aligned}$$

 I_n $h\rightarrow 0$

$$\frac{I-I_n}{h^p}\rightarrow C\quad (C\neq 0, \text{定数}),$$

 I_n p h

$$I-T_n\approx -\frac{h^2}{12}[f'(b)-f'(a)],\tag{4.2.15}$$

$$I-S_n\approx -\frac{1}{180}\left(\frac{h}{2}\right)^4[f'''(b)-f'''(a)],\tag{4.2.16}$$

$$I-C_n\approx -\frac{2}{945}\left(\frac{h}{4}\right)^6[f^{(5)}(b)-f^{(5)}(a)].\tag{4.2.17}$$

 h n $1/4$ $1/16$ $1/64$ $[a,b]$ n $n+1$ T_n $n+1$ $2n+1$ $[x_k,x_{k+1}]$

$$x_{k+\frac{1}{2}}=\frac{1}{2}(x_k+x_{k+1})$$

$$\frac{h}{4}[f(x_k) + 2f(x_{k+\frac{1}{2}}) + f(x_{k+1})].$$

$$h = \frac{b-a}{n}$$

$$T_{2n} = \frac{h}{4} \sum_{k=0}^{n-1} [f(x_k) + f(x_{k+1})] + \frac{h}{2} \sum_{k=0}^{n-1} f(x_{k+\frac{1}{2}}),$$

$$T_{2n} = \frac{1}{2}T_n + \frac{h}{2} \sum_{k=0}^{n-1} f(x_{k+\frac{1}{2}}). \quad (4.3.1)$$

$$I = \int_0^1 \frac{\sin x}{x} dx$$

$$f(x) = \frac{\sin x}{x}$$

$$x = 0$$

$$f(0) = 1$$

$$f(1) = 0.841\,470\,9$$

$$T_1 = \frac{1}{2}[f(0) + f(1)] = 0.920\,735\,5.$$

$$f(1/2) = 0.958\,851\,0$$

$$T_2 = \frac{1}{2}T_1 + \frac{1}{2}f\left(\frac{1}{2}\right) = 0.939\,793\,3.$$

$$f(1/4) = 0.989\,615\,8, \quad f(3/4) = 0.908\,851\,6.$$

$$T_4 = \frac{1}{2}T_2 + \frac{1}{4}\left[f\left(\frac{1}{4}\right) + f\left(\frac{3}{4}\right)\right] = 0.944\,513\,5.$$

k

$$n = 2^k$$

k

T_n

$$0.939\,793\,3$$

$$0.944\,513\,5$$

$$9.945\,690\,9$$

$$0.945\,985\,0$$

$$0.946\,059\,6$$

k T_n

0.946 076 9

0.946 081 5

0.946 082 7

0.946 083 0

0.946 083 1

 I

0.946 083 1

 $k = 3$

9.945 690 9

 T_8 T_8

0

0.945 690 9

 k $k = 9$

0.946 083 0

 $k = 10$

0.946 083 1

 $k = 5$

0.946 059 6

 $f(x) = \sin x/x$ $f(0) = 1$ $[0, 1]$ $n = 32$ $T_{32} = 0.946\,058\,560\,962\,768\,067\dots$

0.946 058 6

 T_n h^2

1/4

$$\frac{I - T_{2n}}{I - T_n} \approx \frac{1}{4}.$$

$$I - T_{2n} \approx \frac{1}{3}(T_{2n} - T_n). \tag{4.3.2}$$

$$\begin{array}{l} T_n \\ T_{2n} \\ T_{2n} \\ \frac{T_{2n}}{3} \\ \frac{1}{3}(T_{2n} - T_n) \\ T_{2n} \end{array}$$

$$\overline{T} = T_{2n} + \frac{1}{3}(T_{2n} - T_n) = \frac{4}{3}T_{2n} - \frac{1}{3}T_n \tag{4.3.3}$$

$$\begin{array}{l} T_4 = 0.944\,513\,5 \\ T_8 = 0.945\,690\,9 \\ I = 0.946\,083\,1 \end{array}$$

$$\overline{T} = \frac{4}{3}T_8 - \frac{1}{3}T_4 = 0.946\,083\,3$$

$$\overline{T}$$

$$S_n = \frac{4}{3}T_{2n} - \frac{1}{3}T_n, \tag{4.3.4}$$

$$\begin{array}{l} T_n \\ T_{2n} \\ S_n \\ h^4 \\ 1/16 \end{array}$$

$$\frac{I - S_{2n}}{I - S_n} \approx \frac{1}{16}, \quad I \approx \frac{16}{15}S_{2n} - \frac{1}{15}S_n.$$

$$\begin{array}{l} C_n \\ S_n \\ S_{2n} \\ C_n \end{array}$$

$$C_n = \frac{16}{15}S_{2n} - \frac{1}{15}S_n. \tag{4.3.5}$$

$$R_n = \frac{64}{63}C_{2n} - \frac{1}{63}C_n. \tag{4.3.6}$$

T_n
 S_n
 C_n
 R_n
 k
 k
 T_2^k
 S_2^{k-1}
 C_2^{k-2}
 R_2^{k-3}
 $0.920\,735\,5$
 $0.939\,793\,3$
 $0.946\,145\,9$
 $0.944\,513\,5$
 $0.946\,086\,9$
 $0.946\,083\,0$
 $0.945\,690\,9$
 $0.946\,083\,3$
 $0.946\,083\,1$
 $0.946\,083\,1$
 $R_1 = 0.946\,083\,1$
 T_2^k
 S_2^{k-1}
 C_2^{k-2}
 R_2^{k-3}
 k
 2^k
 2^{k-1}
 2^{k-2}
 2^{k-3}
 T_{2^k}
 $S_{2^{k-1}}$
 $C_{2^{k-2}}$

$R_{2^{k-3}}$
 $16/15$
 $-1/15$
 $f(x) \in C^\infty[a,b]$

$$T(h) = I + a_1 h^2 + a_2 h^4 + a_3 h^6 + \cdots + a_k h^{2k} + \cdots , \tag{4.3.7}$$

a_k
 $k = 1, 2, \cdots$
 h

$$T\left(\frac{h}{2}\right) = I + \frac{\alpha_1}{4} h^2 + \frac{\alpha_2}{16} h^4 + \frac{\hat{\alpha}_3}{64} h^6 + \cdots . \tag{4.3.8}$$

$\hat{\alpha}_k$
 β_k
 γ_k
 h

$$T_1(h) = \frac{4}{3} T\left(\frac{h}{2}\right) - \frac{1}{3} T(h), \tag{4.3.9}$$

h^2

$$T_1(h) = I + \beta_1 h^4 + \beta_2 h^6 + \beta_3 h^8 + \cdots . \tag{4.3.10}$$

$\{T_1(h)\}$

$$T_1\left(\frac{h}{2}\right) = I + \frac{\beta_1}{16} h^4 + \hat{\beta}_2 h^6 + \beta_3 h^8 + \cdots ,$$

$$T_2(h) = \frac{16}{15} T_1\left(\frac{h}{2}\right) - \frac{1}{15} T_1(h),$$

h^4

$$T_2(h) = I + \gamma_1 h^6 + \gamma_2 h^8 + \cdots .$$

$\{T_2(h)\}$

$$T_0(h) = T(h)$$

$$T_m(h)=\frac{4^m}{4^m-1}T_{m-1}\left(\frac{h}{2}\right)-\frac{1}{4^m-1}T_{m-1}(h)\tag{4.3.11}$$

$$\begin{array}{l} m\\ m=1,2,\cdots \end{array}$$

$$T_m(h)=I+\delta_1h^{2(m+1)}+\delta_2h^{2(m+2)}+\cdots.\tag{4.3.12}$$

$$\begin{array}{l} T_0^{(k)}\\ k\\ T_m^{(k)}\\ \{T_0^{(k)}\}\\ m \end{array}$$

$$T_m^{(k)}=\frac{4^m}{4^m-1}T_{m-1}^{(k+1)}-\frac{1}{4^m-1}T_{m-1}^{(k)}\quad (k=1,2,\cdots)\tag{4.3.13}$$

$$\begin{array}{l} \alpha_1\\ \alpha_2\\ \hat{\alpha}_3\\ \hat{\alpha}_k\\ T_1(h/2)\\ \hat{\beta}_2\\ \beta_3\\ k=1,2,\cdots\\ T_m^{(0)}\\ k=0\\ f\in C^\infty[a,b]\\ T(h)\\ h\rightarrow 0\\ T \end{array}$$

$$\begin{array}{ccccccc} & & & & & & T_0^{(0)} \\ & & & & & & T_0^{(1)} & T_1^{(0)} \\ & & & & & & T_0^{(2)} & T_1^{(1)} & T_2^{(0)} \\ & & & & & & T_0^{(3)} & T_1^{(2)} & T_2^{(1)} & T_3^{(0)} \\ & & & & & & \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

$$\begin{array}{l} f(x) \\ T \\ I \end{array}$$

$$\lim_{k\rightarrow\infty} T_m^{(k)} = I \quad (m \text{ 固定}), \quad \lim_{m\rightarrow\infty} T_m^{(0)} = I.$$

$$\begin{array}{l} T \\ T_0^{(0)} = \frac{b-a}{2}[f(a)+f(b)] \\ 1\rightarrow k \\ k \\ T_0^{(k)} \\ T \\ k+1 \\ T_j^{(k-j)} \\ j=1,2,\cdots,k \\ \varepsilon \\ |T_k^{(0)}-T_{k-1}^{(0)}|<\varepsilon \\ T_k^{(0)} \\ k+1\rightarrow k \\ f(x) \\ [x_k,x_{k+1}] \\ x_{k+\frac{1}{2}}=\frac{1}{2}(x_k+x_{k+1}) \end{array}$$

$$\begin{aligned} f(x) = & f_{k+\frac{1}{2}} + (x-x_{k+\frac{1}{2}})f'_{k+\frac{1}{2}} + \frac{(x-x_{k+\frac{1}{2}})^2}{2!}f''_{k+\frac{1}{2}} + \frac{(x-x_{k+\frac{1}{2}})^3}{3!}f'''_{k+\frac{1}{2}} \\ & + \frac{(x-x_{k+\frac{1}{2}})^4}{4!}f^{(4)}_{k+\frac{1}{2}} + \frac{(x-x_{k+\frac{1}{2}})^5}{5!}f^{(5)}_{k+\frac{1}{2}} + \frac{(x-x_{k+\frac{1}{2}})^6}{6!}f^{(6)}_{k+\frac{1}{2}} + \cdots, \end{aligned} \tag{4.3.14}$$

$$\begin{array}{c}
f_{k+\frac{1}{2}}^{(j)} \\
f^{(j)}(x_{k+\frac{1}{2}}) \\
f(x_k) \\
f(x_{k+1})
\end{array}$$

$$\frac{h}{2}[f(x_k)+f(x_{k+1})] = hf_{k+\frac{1}{2}} + \frac{h}{2!} \left(\frac{h}{2}\right)^2 f''_{k+\frac{1}{2}} + \frac{h}{4!} \left(\frac{h}{2}\right)^4 f^{(4)}_{k+\frac{1}{2}} + \frac{h}{6!} \left(\frac{h}{2}\right)^6 f^{(6)}_{k+\frac{1}{2}} + \cdots ,$$

$$\begin{aligned}
T(h) &= \frac{h}{2} \sum [f(x_k) + f(x_{k+1})] \\
&= h \sum f_{k+\frac{1}{2}} + \frac{h^3}{2! \times 2^2} \sum f''_{k+\frac{1}{2}} + \frac{h^5}{4! \times 2^4} \sum f^{(4)}_{k+\frac{1}{2}} + \frac{h^7}{6! \times 2^6} \sum f^{(6)}_{k+\frac{1}{2}} + \cdots .
\end{aligned}$$

(4.3.15)